A1 Q2c

Edward Tamurkov 20401544

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Showing weak stationarity

c) Show mean does not dep. on t

$E[X_t] = E[Z_0 S_t] + E[Z_1 S_{t-1}]$

$= S_t E[Z_0] + S_{t-1} E[Z_1]$

$= 0 - 0$

$= 0$

$\therefore$ constant $\forall t$ ✓

Show $\gamma(h)$ does not dep. on t

$\gamma(h) = E[X_t X_{t+h}]$ (same reasoning as a), b))

$= E[[Z_0 S_t + Z_1 S_{t-1}][Z_0 S_{t+h} + Z_1 S_{t-1+h}]]$

$= E[Z_0^2 S_t S_{t+h} + Z_0 Z_1 S_t S_{t-1+h} + Z_0 Z_1 S_{t-1} S_{t+h} + Z_1^2 S_{t-1} S_{t-1+h}]$

$(E[Z_0 Z_1] = 0,\ E[Z_0^2] = E[Z_1^2] = \sigma^2$ by def'n)

$= \sigma^2 S_t S_{t+h} + \sigma^2 S_{t-1} S_{t-1+h}$

$= \sigma^2 [S_t S_{t+h} + S_{t-1} S_{t-1+h}]$

We have two cases to consider.

Consider h is even.

$t + h = t$ b/c period = 2

$\Rightarrow \sigma^2 [S_t S_t + S_{t-1} S_{t-1}]$

$\Rightarrow \sigma^2 [S_t^2 + S_{t-1}^2]$

But given period = 2, S_t and S_{t-1} will always have the same value. Let $a = S_t^2 + S_{t-1}^2$ for some $a \in \mathbb{R}$.

$\Rightarrow \sigma^2 a$ for some constant a.

Now, consider h is odd

t and t+h are different parity. Thus, S_t and S_{t+h} are different values

Thus, let $a = S_t S_{t+h}$ and $b = S_{t-1} S_{t-1+h}$ for constants $a, b \in \mathbb{R}$

$\Rightarrow \sigma^2 [a + b]$

We can do this b/c a and b each cover the entire period (S_t even, S_{t+h} odd).

$\therefore$ We have shown in all cases that $\gamma(h)$ does not depend on t given the parity produces some constants $a, b \in \mathbb{R}$ across the period of 2.

$\therefore$ We have shown X_t is weakly stationary.

■

Computing ACF

$$\gamma(h) = \begin{cases} \sigma^2 a & \text{for even } h \\ \sigma^2 (a+b) & \text{for odd } h \end{cases}$$ as per above

$\gamma(0) = \sigma^2 [S_t^2 + S_{t-1}^2] = \sigma^2 a$ as per above

$$\Rightarrow \rho(h) = \begin{cases} \frac{\sigma^2 a}{\sigma^2 a} & \text{for an even } h \\ \frac{\sigma^2(a+b)}{\sigma^2 a} & \text{for an odd } h \end{cases}$$

$$\therefore \rho(h) = \begin{cases} 1 & \text{for an even } h \\ \frac{a+b}{a} & \text{for an odd } h \end{cases} \quad \text{is the ACF of } X_t.$$